

Database Management Systems

Lab – 1

▪ Data Definition Language (DDL) Commands: -

1. CREATE

2. ALTER

a) ADD

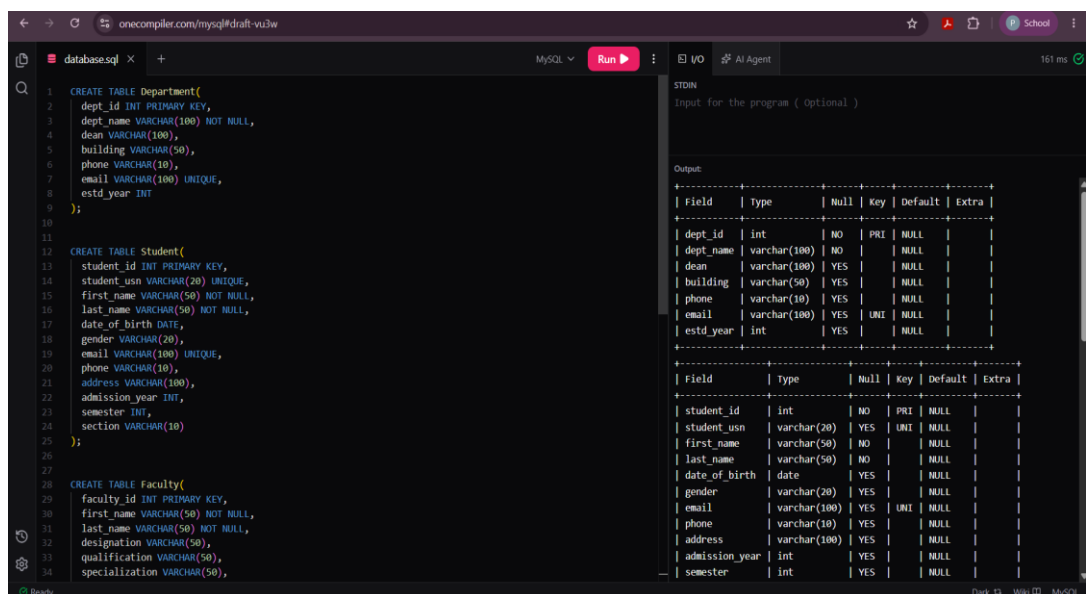
b) MODIFY

c) RENAME

d) DROP

3. DROP

➤ CREATE –



```
1 CREATE TABLE Department(  
2   dept_id INT PRIMARY KEY,  
3   dept_name VARCHAR(100) NOT NULL,  
4   dean VARCHAR(100),  
5   building VARCHAR(50),  
6   phone VARCHAR(10),  
7   email VARCHAR(100) UNIQUE,  
8   estd_year INT  
9 );  
10  
11  
12 CREATE TABLE Student(  
13   student_id INT PRIMARY KEY,  
14   student_usn VARCHAR(20) UNIQUE,  
15   first_name VARCHAR(50) NOT NULL,  
16   last_name VARCHAR(50) NOT NULL,  
17   date_of_birth DATE,  
18   gender VARCHAR(20),  
19   email VARCHAR(100) UNIQUE,  
20   phone VARCHAR(10),  
21   address VARCHAR(100),  
22   admission_year INT,  
23   semester INT,  
24   section VARCHAR(10)  
25 );  
26  
27  
28 CREATE TABLE Faculty(  
29   faculty_id INT PRIMARY KEY,  
30   first_name VARCHAR(50) NOT NULL,  
31   last_name VARCHAR(50) NOT NULL,  
32   designation VARCHAR(50),  
33   qualification VARCHAR(50),  
34   specialization VARCHAR(50),
```

Output:

Field	Type	Null	Key	Default	Extra
dept_id	int	NO	PRI	NULL	
dept_name	varchar(100)	NO		NULL	
dean	varchar(100)	YES		NULL	
building	varchar(50)	YES		NULL	
phone	varchar(10)	YES		NULL	
email	varchar(100)	YES	UNI	NULL	
estd_year	int	YES		NULL	

Field	Type	Null	Key	Default	Extra
student_id	int	NO	PRI	NULL	
student_usn	varchar(20)	YES	UNI	NULL	
first_name	varchar(50)	NO		NULL	
last_name	varchar(50)	NO		NULL	
date_of_birth	date	YES		NULL	
gender	varchar(20)	YES		NULL	
email	varchar(100)	YES	UNI	NULL	
phone	varchar(10)	YES		NULL	
address	varchar(100)	YES		NULL	
admission_year	int	YES		NULL	
semester	int	YES		NULL	

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database.sql x + MySQL Run VO AI Agent 161 ms

```

28 CREATE TABLE Faculty(
35   email VARCHAR(100) UNIQUE NOT NULL,
36   phone VARCHAR(15),
37   dept_id INT NOT NULL,
38   hire_date DATE,
39   salary DECIMAL(10, 2),
40   FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
41 );
42
43
44 CREATE TABLE Course(
45   course_id INT PRIMARY KEY,
46   course_code VARCHAR(20) NOT NULL UNIQUE,
47   course_name VARCHAR(50),
48   course_type VARCHAR(10),
49   credits INT NOT NULL,
50   semester INT,
51   faculty_id INT,
52   schedule VARCHAR(50),
53   FOREIGN KEY (faculty_id) REFERENCES Faculty(faculty_id)
54 );
55
56
57 CREATE TABLE Enrollment(
58   enrollment_id INT PRIMARY KEY,
59   enrollment_date DATE NOT NULL,
60   student_id INT NOT NULL,
61   course_id INT NOT NULL,
62   academic_year INT,
63   FOREIGN KEY (student_id) REFERENCES Student(student_id),
64   FOREIGN KEY (course_id) REFERENCES Course(course_id),
65   UNIQUE(student_id, course_id)
66 );
67
68
69 Ready

```

STDIN

Input for the program (optional)

Output:

Field	Type	Null	Key	Default	Extra
faculty_id	int	NO	PRI	NULL	
first_name	varchar(50)	NO		NULL	
last_name	varchar(50)	NO		NULL	
designation	varchar(50)	YES		NULL	
qualification	varchar(50)	YES		NULL	
specialization	varchar(50)	YES		NULL	
email	varchar(100)	NO	UNI	NULL	
phone	varchar(15)	YES		NULL	
dept_id	int	NO	MUL	NULL	
hire_date	date	YES		NULL	
salary	decimal(10,2)	YES		NULL	

Field	Type	Null	Key	Default	Extra
course_id	int	NO	PRI	NULL	
course_code	varchar(20)	NO	UNI	NULL	
course_name	varchar(50)	YES		NULL	
course_type	varchar(10)	YES		NULL	
credits	int	NO		NULL	
semester	int	YES		NULL	
faculty_id	int	YES	MUL	NULL	
schedule	varchar(50)	YES		NULL	

Dark Wiki MySQL

```

42
43
44 CREATE TABLE Course(
45   course_id INT PRIMARY KEY,
46   course_code VARCHAR(20) NOT NULL UNIQUE,
47   course_name VARCHAR(50),
48   course_type VARCHAR(10),
49   credits INT NOT NULL,
50   semester INT,
51   faculty_id INT,
52   schedule VARCHAR(50),
53   FOREIGN KEY (faculty_id) REFERENCES Faculty(faculty_id)
54 );
55
56
57 CREATE TABLE Enrollment(
58   enrollment_id INT PRIMARY KEY,
59   enrollment_date DATE NOT NULL,
60   student_id INT NOT NULL,
61   course_id INT NOT NULL,
62   academic_year INT,
63   FOREIGN KEY (student_id) REFERENCES Student(student_id),
64   FOREIGN KEY (course_id) REFERENCES Course(course_id),
65   UNIQUE(student_id, course_id)
66 );
67
68
69 DESC Department;
70 DESC Student;
71 DESC Faculty;
72 DESC Course;
73 DESC Enrollment;
74
75 Ready

```

Output:

Field	Type	Null	Key	Default	Extra
dept_id	int	NO	MUL	NULL	
hire_date	date	YES		NULL	
salary	decimal(10,2)	YES		NULL	

Field	Type	Null	Key	Default	Extra
course_id	int	NO	PRI	NULL	
course_code	varchar(20)	NO	UNI	NULL	
course_name	varchar(50)	YES		NULL	
course_type	varchar(10)	YES		NULL	
credits	int	NO		NULL	
semester	int	YES		NULL	
faculty_id	int	YES	MUL	NULL	
schedule	varchar(50)	YES		NULL	

Field	Type	Null	Key	Default	Extra
enrollment_id	int	NO	PRI	NULL	
enrollment_date	date	NO		NULL	
student_id	int	NO	MUL	NULL	
course_id	int	NO	MUL	NULL	
academic_year	int	YES		NULL	

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➤ ALTER –

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Upgrade

AIMySQLRun

database.sql

```
48 CREATE TABLE Course(
49 );
50
51 ALTER TABLE Department
52 ADD COLUMN website VARCHAR(100);
53
54 ALTER TABLE Department
55 DROP COLUMN estd_year;
56
57 ALTER TABLE Student
58 MODIFY phone VARCHAR(15);
59
60 ALTER TABLE Faculty
61 CHANGE COLUMN specialization subject_specialization VARCHAR(50);
62
63 ALTER TABLE Department
64 DROP email;
65
66 ALTER TABLE Student
67 ADD COLUMN blood_group VARCHAR(5);
68
69 ALTER TABLE Faculty
70 ADD COLUMN experience_years INT NOT NULL DEFAULT 0;
71
72 ALTER TABLE Course
73 RENAME TO Courses;
74
75 SHOW TABLES;
76
77 DESC Department;
78 DESC Student;
79 DESC Faculty;
80 DESC Courses;
```

AI Agent

344 ms

STDOUT

Input for the program (optional)

Output:

Tables in sandbox db |

courses

department

faculty

student

Field | Type | Null | Key | Default | Extra |

dept_id | int | NO | PRI | NULL | |

dept_name | varchar(100) | NO | | NULL | |

dean | varchar(100) | YES | | NULL | |

building | varchar(50) | YES | | NULL | |

phone | varchar(10) | YES | | NULL | |

website | varchar(100) | YES | | NULL | |

Field | Type | Null | Key | Default | Extra |

student_usn | int | NO | PRI | NULL | |

first_name | varchar(50) | NO | | NULL | |

last_name | varchar(50) | NO | | NULL | |

Dark Wiki MySQL

Output:

Field | Type | Null | Key | Default | Extra |

student_usn | int | NO | PRI | NULL | |

first_name | varchar(50) | NO | | NULL | |

last_name | varchar(50) | NO | | NULL | |

date_of_birth | date | YES | | NULL | |

gender | varchar(20) | YES | | NULL | |

email | varchar(100) | YES | UNI | NULL | |

phone | varchar(15) | YES | | NULL | |

address | varchar(100) | YES | | NULL | |

admission_year | int | YES | | NULL | |

semester | int | YES | | NULL | |

section | varchar(10) | YES | | NULL | |

blood_group | varchar(5) | YES | | NULL | |

Field | Type | Null | Key | Default | Extra |

faculty_id | int | NO | PRI | NULL | |

first_name | varchar(50) | NO | | NULL | |

last_name | varchar(50) | NO | | NULL | |

designation | varchar(50) | YES | | NULL | |

qualification | varchar(50) | YES | | NULL | |

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	qualification		varchar(50)		YES				NULL			
	subject_specialization		varchar(50)		YES				NULL			
	email		varchar(100)		NO		UNI		NULL			
	phone		varchar(15)		YES				NULL			
	dept_id		int		NO		MUL		NULL			
	hire_date		date		YES				NULL			
	salary		decimal(10,2)		YES				NULL			
	experience_years		int		NO				0			
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+												
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+												
	Field		Type		Null		Key		Default		Extra	
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+												
	course_code		int		NO		PRI		NULL			
	course_name		varchar(50)		YES				NULL			
	course_type		varchar(10)		YES				NULL			
	credits		int		NO				NULL			
	semester		int		YES				NULL			
	faculty_id		int		YES		MUL		NULL			
	schedule		varchar(50)		YES				NULL			
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+												

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➤ DROP –

OneCompiler

Upgrade

AI

MySQL

Run

...

database.sql

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```

30
31 CREATE TABLE DepartmentClub(
32   club_id INT PRIMARY KEY,
33   club_name VARCHAR(100),
34   description VARCHAR(200),
35   presedint VARCHAR(100),
36   vice_president VARCHAR(100),
37   coordinator VARCHAR(100),
38   established_year INT,
39   contact_email VARCHAR(100) UNIQUE
40 );
41
42 ALTER TABLE Department
43 ADD COLUMN website VARCHAR(100);
44
45 ALTER TABLE Department
46 DROP COLUMN estd_year;
47
48 ALTER TABLE Student
49 MODIFY phone VARCHAR(15);
50
51 ALTER TABLE Faculty
52 CHANGE COLUMN specialization subject_specialization VARCHAR(50);
53
54 ALTER TABLE Department
55 DROP email;
56
57 ALTER TABLE Student
58 ADD COLUMN blood_group VARCHAR(5);
59
60 ALTER TABLE Faculty
61 ADD COLUMN experience_years INT NOT NULL DEFAULT 0;
62

```

362 ms

AI Agent

Stdin

Input for the program (Optional)

Output:

	Field		Type		Null		Key		Default		Extra	
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+												
	course_code		int		NO		PRI		NULL			
	course_name		varchar(50)		YES				NULL			
	course_type		varchar(10)		YES				NULL			
	credits		int		NO				NULL			
	semester		int		YES				NULL			
	faculty_id		int		YES		MUL		NULL			
	schedule		varchar(50)		YES				NULL			
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+												
	Field		Type		Null		Key		Default		Extra	
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+												
	club_id		int		NO		PRI		NULL			
	club_name		varchar(100)		YES				NULL			
	description		varchar(200)		YES				NULL			
	presedint		varchar(100)		YES				NULL			
	vice_president		varchar(100)		YES				NULL			
	coordinator		varchar(100)		YES				NULL			
	established_year		int		YES				NULL			
	contact_email		varchar(100)		YES		UNI		NULL			
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+												

Dark

Wiki

MySQL

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database.sql × +

```
51 CREATE TABLE DepartmentClub(  
52 );  
53  
54  
55 ALTER TABLE Department  
56 ADD COLUMN website VARCHAR(100);  
57  
58 ALTER TABLE Department  
59 DROP COLUMN estd_year;  
60  
61 ALTER TABLE Student  
62 MODIFY phone VARCHAR(15);  
63  
64 ALTER TABLE Faculty  
65 CHANGE COLUMN specialization subject_specialization VARCHAR(50);  
66  
67 ALTER TABLE Department  
68 DROP email;  
69  
70 ALTER TABLE Student  
71 ADD COLUMN blood_group VARCHAR(5);  
72  
73 ALTER TABLE Faculty  
74 ADD COLUMN experience_years INT NOT NULL DEFAULT 0;  
75  
76 ALTER TABLE Course  
77 RENAME TO Courses;  
78  
79 SHOW TABLES;  
80  
81 DESC Department;  
82  
83 DESC Student;  
84  
85 DESC Faculty;  
86  
87 DESC Courses;
```

AI Agent 363 ms

STDIN
Input for the program (Optional)

Output:

```
+-----+  
| Tables_in_sandbox_db |  
+-----+  
| courses                |  
| department             |  
| departmentclub         |  
| faculty                |  
| student                |  
+-----+
```

Field	Type	Null	Key	Default	Extra
dept_id	int	NO	PRI	NULL	
dept_name	varchar(100)	NO		NULL	
dean	varchar(100)	YES		NULL	
building	varchar(50)	YES		NULL	
phone	varchar(10)	YES		NULL	
website	varchar(100)	YES		NULL	

Field	Type	Null	Key	Default	Extra
student_usn	int	NO	PRI	NULL	
first_name	varchar(50)	NO		NULL	

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database.sql × +

```
51  
52 ALTER TABLE Department  
53 ADD COLUMN website VARCHAR(100);  
54  
55 ALTER TABLE Department  
56 DROP COLUMN estd_year;  
57  
58 ALTER TABLE Student  
59 MODIFY phone VARCHAR(15);  
60  
61 ALTER TABLE Faculty  
62 CHANGE COLUMN specialization subject_specialization VARCHAR(50);  
63  
64 ALTER TABLE Department  
65 DROP email;  
66  
67 ALTER TABLE Student  
68 ADD COLUMN blood_group VARCHAR(5);  
69  
70 ALTER TABLE Faculty  
71 ADD COLUMN experience_years INT NOT NULL DEFAULT 0;  
72  
73 ALTER TABLE Course  
74 RENAME TO Courses;  
75  
76 DROP TABLE DepartmentClub;  
77  
78 SHOW TABLES;  
79  
80 DESC Department;  
81  
82 DESC Student;  
83  
84 DESC Faculty;  
85  
86 DESC Courses;
```

AI Agent Error

STDIN
Input for the program (Optional)

Output:

```
+-----+  
| Tables_in_sandbox_db |  
+-----+  
| courses                |  
| department             |  
| departmentclub         |  
| faculty                |  
| student                |  
+-----+
```

Field	Type	Null	Key	Default	Extra
dept_id	int	NO	PRI	NULL	
dept_name	varchar(100)	NO		NULL	
dean	varchar(100)	YES		NULL	
building	varchar(50)	YES		NULL	
phone	varchar(10)	YES		NULL	
website	varchar(100)	YES		NULL	

Field	Type	Null	Key	Default	Extra
student_usn	int	NO	PRI	NULL	
first_name	varchar(50)	NO		NULL	

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